MCA Software Engineering — Tutorial 8

Q1. Counting module dependencies (5 min; 3 marks)

A software system has five modules: Login (L), Catalogue (C), Loan (O), Fine (F), and Reports (R). The directed calls are O→C, O→L, O→F, F→O, and R→O. (a) Count all directed dependencies. (b) Find out-degree of O (number of modules O calls) and in-degree of O (number calling O). (c) State one way to avoid an unnecessary dependency.

Answer / working:

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Q2. Change in coupling (8 min; 4 marks)

Before redesign, 8 modules have 18 directed inter-module calls. After redesign, they have 12. For this question, define coupling density = observed directed calls ÷ [n(n−1)], where n is the number of modules. Calculate (a) the density before, (b) the density after, and (c) percentage reduction in the number of calls. Explain whether the redesign reduced this measure of coupling.

Answer / working:

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Q3. Cohesion of responsibilities (9 min; 4 marks)

A module contains 10 tasks. Six tasks directly support its main purpose; the other four are unrelated. For this exercise only, focus ratio = relevant tasks ÷ total tasks × 100%. (a) Calculate the original ratio. (b) Move three unrelated tasks to suitable modules; calculate the ratio for the remaining seven tasks. (c) Explain why the move improves focus.

Answer / working:

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Q4. Interface requests (10 min; 5 marks)

In a student portal, Course module receives 150 enrollment checks/hour from Enrollment module and 90 checks/hour from Timetable module. Each request sends 200 bytes and receives 100 bytes. Assume 1 kB = 1000 bytes. (a) Total requests/hour? (b) Total exchanged bytes/hour including requests and responses? (c) Total kB/hour? (d) Give one benefit of using this single clear interface.

Answer / working:

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Q5. Choose a better modular design (13 min; 4 marks)

Two designs provide the same requirements. Design A uses 6 modules and 14 directed inter-module dependencies. Design B uses 6 modules and 9 directed dependencies. Use density = calls ÷ [n(n−1)]. (a) Calculate both densities.

(b) By what percent did the number of dependencies fall from A to B? (c) Is lower density alone enough to prove B is better? State one additional check (for example, cohesion or correctness).

Answer / working:

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List of Formulas:

Counting module dependencies

Total dependencies (D) = number of directed calls between different modules.

Out-degree of module M = number of outgoing directed calls from M.

In-degree of module M = number of incoming directed calls to M.

Change in coupling

Coupling density (CD) = D / [n × (n − 1)]

Percentage reduction = [(D_before − D_after) / D_before] × 100%

D = observed directed inter-module calls; n = number of modules.

Cohesion of responsibilities

Focus ratio (%) = (Relevant tasks / Total tasks) × 100

When unrelated tasks are moved out, recalculate both relevant and total tasks for the remaining module.

Relevant tasks = tasks supporting the module's main purpose.

Interface requests

Total requests/hour = sum of requests/hour from all calling modules.

Bytes/hour = Total requests/hour × (Request size + Response size).

kB/hour = Bytes/hour / 1000 (as defined in this exercise).

Comparing two designs

Density for each design = D / [n × (n − 1)]

Reduction from A to B (%) = [(D_A − D_B) / D_A] × 100%

Compare densities only when using the same definition; check cohesion, clarity and maintainability too.

Quick symbol reference

Symbol	Meaning
D	Directed dependencies / inter-module calls
n	Number of modules
CD	Coupling density
D_before, D_after	Dependencies before and after redesign
D_A, D_B	Dependencies in design A and design B

Student tip: Write the formula, substitute the given numbers, calculate the answer, and include units where appropriate. High cohesion and low unnecessary coupling are desirable design goals.